Back Error Propogation

TODO: create an actual diagram instead of this thing



For some given input, the network should produce the output:

$$\begin{bmatrix} t_0 \\ t_1 \end{bmatrix}$$

Let E be the error function for the network. It there for must be dependent on the output of the network and the target values. In this case, we use mean squared error (MSE).

$$E = E\left(\begin{bmatrix} a_4 \\ a_5 \end{bmatrix}, \begin{bmatrix} t_0 \\ t_1 \end{bmatrix}\right)$$

$$E = \frac{1}{2} * \left[(t_0 - a_4)^2 + (t_1 - a_5)^2\right]$$

To calculate the value of any given node:

$$\text{output of node } k = f\left(\sum_{i=0}^{n} w_i a_i\right)$$

where f is the sigmoid activation function

$$f(x) = \frac{1}{1 + e^{-}k}$$

$$f'(x) = f(x)(1 - f(x))$$

Now we know how to do a forward pass i.e. how to calculate the output of the network for some given input. Currently, the network will produce a useless random output. We want to train the network to by adjusting the weights of the network.

How much to change any given node k:

$$\Delta w_k = \delta_k a_i$$

$$w_k = w_k + \eta \Delta w_k$$

Δw_k	change in the weight k
η	learning rate (a constant/hyper parameter)
δ_k	the ending node's error
a_i	the incoming node's output

How did we come up with this formula?

For a node on the output layer

To figure out how much a give node contributes to the network's error.

$$\delta_k = \frac{\partial E}{\partial S_k} = \frac{\partial E}{\partial a_k} \cdot \frac{\partial a_k}{\partial S_k}$$

For any node in the output layer $k \in \{4, 5\}$:

$$\frac{\partial E}{\partial a_k} = \frac{\partial}{\partial a_k}\left(\frac{1}{2} * \left[(t_0 - a_4)^2 + (t_1 - a_5)^2\right]\right)$$

eg) Let $k = 4$

$$\frac{\partial E}{\partial a_4} = \frac{1}{2} * 2(t_0 - a_4) \cdot (-1) = \boxed{t_0 - a_4}$$

and the second part of the equation

$$\frac{\partial a_k}{\partial S_k} = \frac{\partial}{\partial S_k}(f(S_k)) = \boxed{f'(S_k)}$$

Substituting both parts back in

$$\delta_k = \frac{\partial E}{\partial a_k} \cdot \frac{\partial a_k}{\partial S_k} = \boxed{(a_k - t_{k+4}) f'(S_k)}$$

$$\Delta w_k = \frac{\partial E}{\partial w_k} = \underbrace{\frac{\partial E}{\partial a_k} \cdot \frac{\partial a_k}{\partial S_k}}_{\delta_k} \cdot \frac{\partial S_k}{\partial w_k}$$

We just solved for δ_k, but we still need to find $\frac{\partial S_k}{\partial w_k}$.

$$\frac{\partial S_k}{\partial w_k} = \frac{\partial}{\partial w_k}\left(\sum_{i=0}^{n} w_i a_i\right)$$

$$= \frac{\partial}{\partial w_k}(w_1 a_1 + w_2 a_2 + ... + w_n a_n)$$

$$= \frac{\partial}{\partial w_k}(w_1 a_1) + \frac{\partial}{\partial w_k}(w_2 a_2) + ... + \frac{\partial}{\partial w_k}(w_n a_n)$$

$$= 0 + 0 + ... + \frac{\partial}{\partial w_k}(w_k a_k) + ... + 0 = a_k$$

$$\boxed{\frac{\partial S_k}{\partial w_k} = a_k}$$

$$\Delta w_k = \frac{\partial E}{\partial w_k} = \frac{\partial E}{\partial a_k} \cdot \frac{\partial a_k}{\partial S_k} \cdot \frac{\partial S_k}{\partial w_k}$$

$$\Delta w_k = \left(t_{k_1} - a_{k_1}\right)\left(f'\left(S_{k_1}\right)\right)\left(a_{k_2}\right)$$

$$w_k = w_k + \eta \Delta w_k$$

For a weight going to and from arbitrary nodes

$$E(\text{the whole network}) = E(a_4) + E(a_5)$$

$$= \sum_{i=\text{out}} E(a_i), \text{where out is the set of output indices } \{4, 5\} \text{ in this network}$$

For an arbitrary weight w_k where $0 \leq k \leq n$.

$$\frac{\partial E}{\partial w_k} = \frac{\partial}{\partial w_k}(E(a_4) + E(a_5)) = \frac{\partial E(a_4)}{\partial w_k} + \frac{\partial E(a_5)}{\partial w_k}$$

Solving for $\frac{\partial E(a_4)}{\partial w_k}$ first, you will need:

$$a_4 = f(S_4) \qquad E(a_4) = (t_0 - a_4) \qquad S_4 = \sum_{i=0}^{n} w_i a_i$$

$$\frac{\partial}{\partial w_k}(E(a_4))$$

$$= \frac{\partial}{\partial w_k}(t_0 - a_4)$$

$$= \frac{\partial}{\partial w_k}(t_0 - f(S_4))$$

$$= \frac{\partial}{\partial w_k}\left(t_0 - f\left(\sum_{i=0}^{n} w_i a_i\right)\right)$$

$$= \frac{\partial}{\partial w_k}\left(t_0 - f\left(\sum_{i=0}^{n} w_i a_i\right)\right)$$

$$= \frac{\partial}{\partial w_k}(t_0 - f(w_0 a_0 + w_1 a_1 + ... + w_k a_k + ... + w_n a_n))$$

$$= \frac{\partial}{\partial w_k}(t_0) - \frac{\partial}{\partial w_k} f(w_0 a_0 + w_1 a_1 + ... + w_k a_k + ... + w_n a_n)$$

Derivative of t_0 w.r.t. w_k is the derivative of a constant

$$= 0 - \frac{\partial}{\partial w_k} f(w_0 a_0 + w_1 a_1 + ... + w_k a_k + ... + w_n a_n)$$

Applying the chain rule

$$= -f'(w_0 a_0 + w_1 a_1 + ... + w_k a_k + ... + w_n a_n) \cdot \frac{\partial}{\partial w_k}(w_0 a_0 + w_1 a_1 + ... + w_k a_k + ... + w_n a_n)$$

$$= -f'(w_0 a_0 + w_1 a_1 + ... + w_k a_k + ... + w_n a_n) \cdot (0 + 0 + ... + a_k + ... + 0)$$

$$= -f'(S_4) \cdot a_k$$